IPO 2016 - procedures/software

IPO 2016 - Software

For IPO2016 we used a collection of tools to automate the registration/writing/evaluation/administration process.

Student app

- Upload and confirm essay pdf

Essay evaluation app

- collect scores of readers
- assigns papers to readers

Administration app

- Student/teacher import (output from caldera form)
- Update teacher/student data
- Reading group import
- Essay overview
- Monitor evaluation process

Technology

Webserver

- Debian Linux
- PHP application
- MySQL database
- PDFTK to add headers to essay pdf's
- Samba to share pdf output

Batch printing

- watchDirectory (http://www.watchdirectory.net/)

IPO 2016 - Online registrations

Software

- IPO2016 website is a WordPress site
- We used Caldera Forms plugin for the registration process.
  After the registration deadline, all data was imported into the MySQL database

Teacher data

- country
- first name
- last name
- e-mail
- school
- gender (m/f)
- languages (English/French/German/Spanish)
- meal preference (no preference/vegetarian/vegan)
- special dietary/health/religious needs
- t-shirt size (s/m/l/xl/xxl)
- delegation leader (yes/no)

Student data

- country
- first name
- last name
- e-mail
- school
- gender (m/f)
- date of birth
- essay language (English/French/German/Spanish)
- keyboard preference (qwerty/azerty)
- meal preference (no preference/vegetarian/vegan)
- special dietary/health/religious needs

IPO 2016 - Essay evaluation procedure

We used this procedure for the essay evaluation process

Stage 1

- Each essay is assigned to 4 readers
- Readers read in group, discuss and return scores
- Evaluation:
    - Score distance > 3 then the essay needs an extra reading
    - Score distance <= 3, Average score < 7 then the reading process is done for that essay
    - Score distance <= 3, Average score >= 7 then the essay is allowed for stage 2

Stage 2

- Essays for stage 2 are assigned to 2 new readers
- Readers read individually, return score
- Final essay score = average score of all 6/7 scores

After stage 2, the Jury of the IPO selects as many of the best essays as it wishes for consideration in the third round.

Stage 3

The IPO Steering Board decides on the distribution or prizes.

IPO 2016 - Writing/Reading schedule

09.00 h - Start of Essay Competition
13.00 h - End of Essay Competition

14.00 h - Start Essay Evaluation
18.00 h - End Essay Evaluation

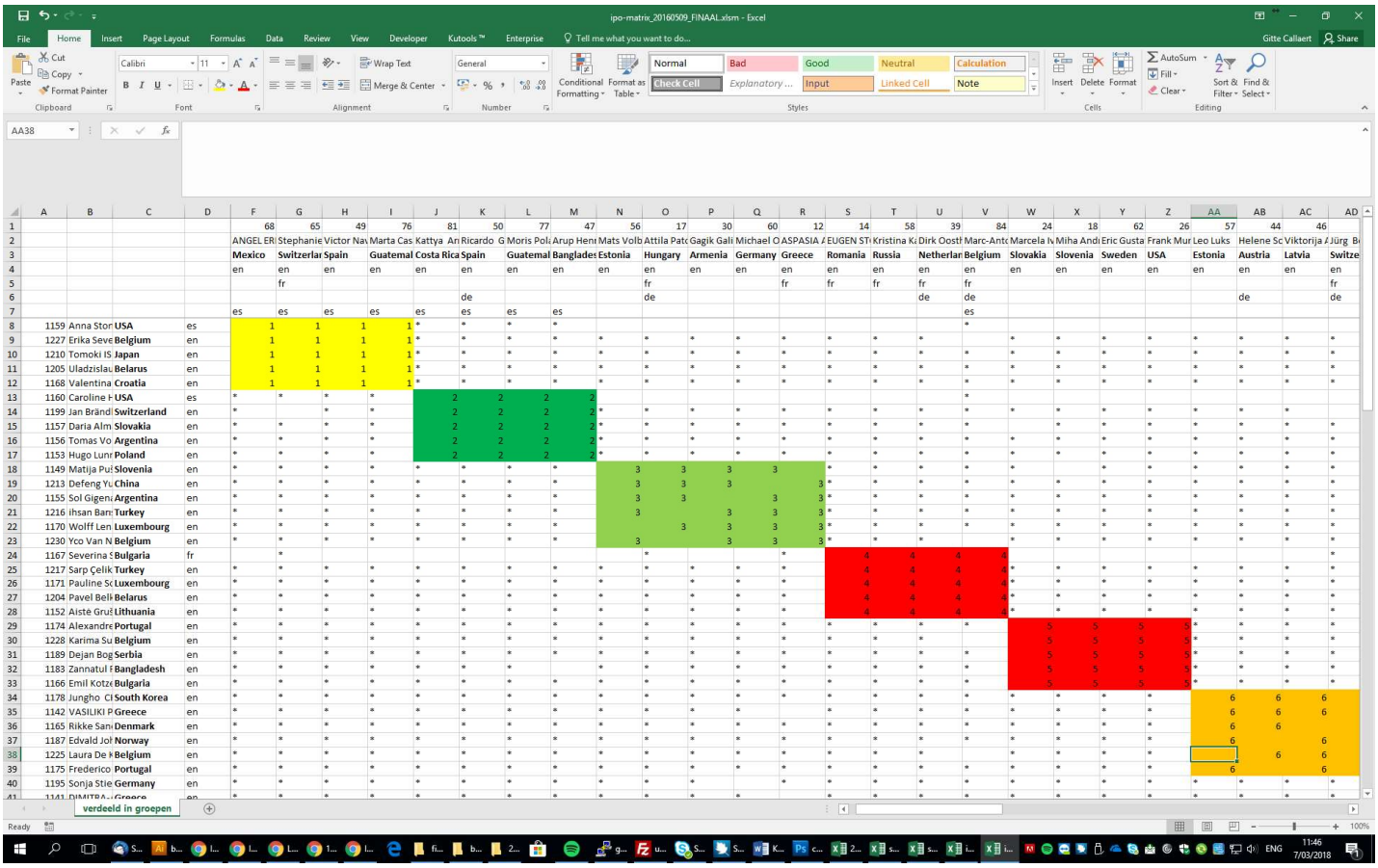
1 hour to print all essays for stage 1
4 hours to evaluate all essays

Data preparation

Stage 1 needs data preparation: find groups of 4 readers that can read the same 4 essays (or 5 essays if necessary)
Limitations:

- Readers can't read papers from their own country
- Readers can only read papers in a valid language

If everyone would write in English, and everyone could read and evaluate English essays, this would be an easy puzzle to solve. We did not implement an automated solution for this problem, we solved this using Microsoft Excel.



Our administration application exports an empty matrix.
Left: a list of all students, including id, country and language
Top: a list of all teachers, including id, country and possible languages they can read

The dots indicate if a teacher can read the essay of the corresponding student. With a few simple row/column operations, we can create groups of matching students/readers.

IPO2016

- 74 readers: 16 groups of 4 + 2 groups of 5 = 18 groups
- 93 essays: each group of 4 teachers get 5 essays; 2 groups of 5 teachers and 1 group of 4 teachers get an extra essay. In group 3 in the example above, each teacher only reads 5 of the six essays assigned to the group.
- Teachers and essays in each group all from different countries

The end result of this matrix is a list of ID's:

Student ID1, Teacher ID1, Teacher ID2, Teacher ID3, Teacher ID4
Student ID2, Teacher ID1, Teacher ID2, Teacher ID3, Teacher ID4
Student ID3, Teacher ID1, Teacher ID2, Teacher ID3, Teacher ID4
Student ID4, Teacher ID1, Teacher ID2, Teacher ID3, Teacher ID4
…

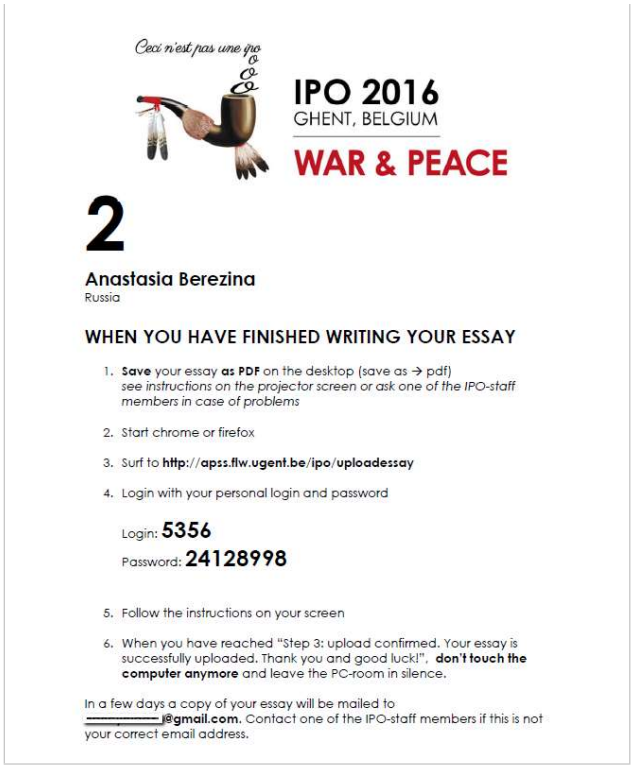
We import this list in our admin application.

Writing process

Preparation

Upon arrival in Ghent, all students received a paper with all details about the writing procedure

- location where the essay writing takes place
- start hour, end hour
- student ID & password for the essay upload application
- essay language
- keyboard type (azerty/qwerty)



Computer rooms

- each computer is switched on, logged on with guest account
- each computer is labelled with a Student Name, Country, Student ID, Essay Language, Keyboard type
- on the computer desktop is a blank Microsoft Word template
- every computer room has 2 people to support the writing process



Writing

- student opens Word template that creates an empty Word document
- template has large empty header, student can't write in that space
  (during the essay evaluation, our software adds extra info in this header space)
- student writes essay and saves the essay as pdf on the computer desktop

Essay upload

1. student logs on to the student application with ID and password
2. student chooses a writing topic and uploads the essay pdf file
3. student gets a preview of the uploaded pdf and confirms the upload
4. after a correct upload, the account is locked

We did not implement an online writing component.

Output

- Each uploaded essay gets an Essay ID
- The upload application renames the uploaded pdf's to their Essay ID
- pdf's are available in a shared folder (samba share) and in the administration application



Printing process

We only had one hour to
- print all essays for stage 1 (4 x 90 essays)
- group all essays per reading group

We decided to print 3 additional copies of each essay
- 1 in case an extra reading is necessary in stage 2
- 2 in case the essay makes it to stage 2

We only had 4 hours to evaluate all essays. We did not want to waste time printing essays on demand for extra reading in stage 2.

Stage 1

We used software to automate the printing process for stage 1.

We wrote a script that generates:
- pdf's per teacher, per group, based on the result of our group matrix. Each pdf gets in the header the essay ID and group number.
- A score sheet per teacher. This sheet has the teacher's name, ID, group number and a list of essay ID's + empty score for all essay's that have to be read by that teacher in stage 1.

We used the software 'watchDirectory' to monitor pdf's on a network share and send the pdf's to multiple printer/copiers.

Stage 2

Essays for extra reading and stage 2 were printed during the first hour of the evaluation process.

Examples

IPO 2016 Essay	Group	Essay	Score
	18	327	/ 10

Aristotle, quotation nr 1

1. Introduction

Aristotle, a Greek, Ancient philosopher, focuses in the quotation on the human ability of expressing the thoughts. Therefore, the quotation finds its place in the vast domain of philosophy called the philosophy of language. Aristotle presents here a vision of the components of the process of expressing the thought and equally looks for some similarities in expressing the thoughts in the whole mankind. In other words he tries to answer the question why people can communicate, if their bodies and languages are different.

The key-words of the quotation are "spoken sounds", "written marks", "affections of the soul" and "actual things". These are components of human possibility to utter some thoughts. Aristotle presents the relation between them as follows:

The base for all the possibility of utterance are atual things, so something that we can talk about. Actual things provoke some thoughts. He called it "affections of the soul". The name seems today really poetic, but it covers as well the idea that these are some perceptions which can affect our soul or our mind. The next element are spoken sounds, when people utter the thought (because we can call the affection of soul a thought) and communicate it so that other people could understand it. The third component are written marks which are symbols of spoken sounds.

This was his vision of the process of expressing the thoughts. The next and essential part of this scheme is the relation between all those components and the whole mankind. Aristotle claims that even if sounds and written marks are different for all the people, the affections of soul are the same and the actual things are shared by everyone. It assumes the realistic interpretation of the world when everybody lives in the one world which is objectively accessible to our inquiry. Consequently, the object of inquiry is not the subjective representation, but something that is univerally accessible. The other even more strong assumption is that people's affections, so thoughts are in fact the same, while they talk about the same things. It is based on the likeness of thoughts to the actual things.

1

Ceci n'est pas une ipo

IPO 2016
GHENT, BELGIUM
WAR & PEACE

Teacher-ID	11
Name	CHRISTINA SELLI
Group	18

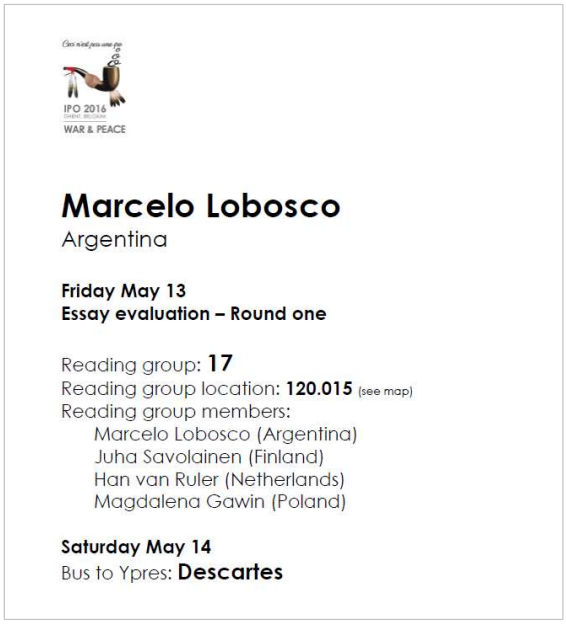
Essay-ID	Score
327	
332	
343	
347	
377	

Evaluation process

Preparation

Upon arrival in Ghent, all teachers received a paper with all details about the reading procedure
- location where essay reading takes place (address, building)
- time schedule
- reading group number for stage 1, reading group location, reading group members (name +

country)



In the reading rooms, all essays and score sheets where available when the readers arrived.

Evaluation application

In order to save time, we decided not to split evaluation stage 1 and 2. When teachers return with their scores in stage 1, we check if it is possible for the reader to read another essay (extra reading for stage 1 or start reading for stage 2). Teachers can choose to accept or decline.

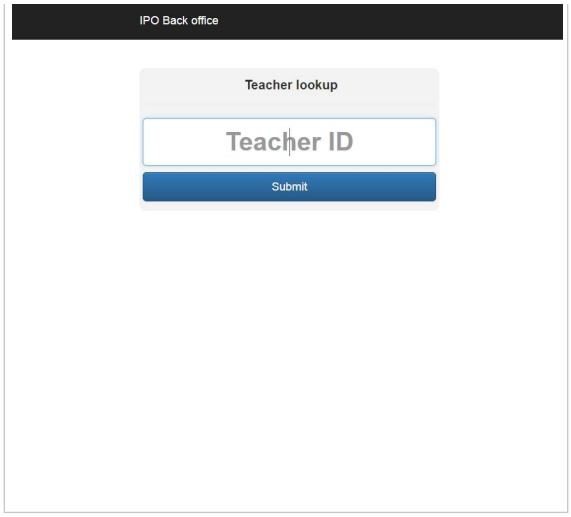
The main idea is: assign essays to teachers for as long as there are essays to be read.

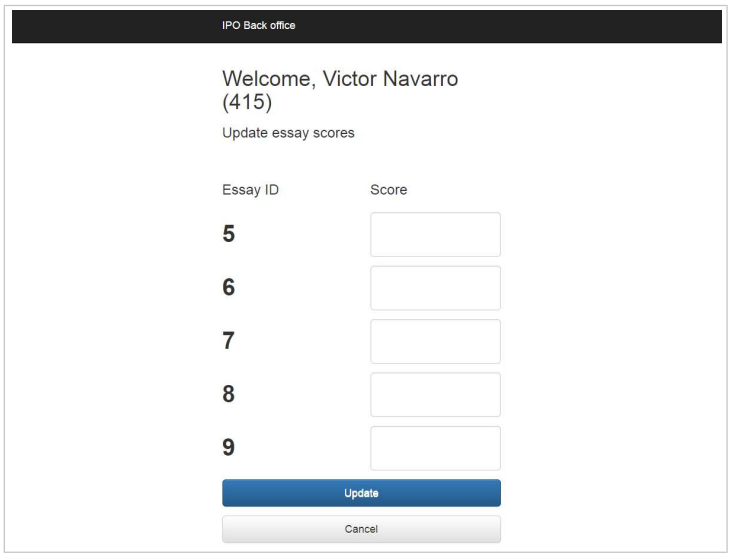
How we implement this:

On each score update/essay assignment, we calculate the status of the essay
- waiting for scores
- needs more reading
- finished

round = 1
- each essay was assigned to 4 teachers
- on each score update, calculate status:
    - no 4 scores => status: waiting for scores
    - 4 scores
        - - score distance > 3 => round = 1, status = needs more reading
        - - average >= 7 => round 2 = 2 , status = needs more reading
        - - average < 7, round = 1, status: finished

round = 2
- essay needs 2 extra readings
    - Assign essay to teacher on score return
- on each score update, calculate status:
    - no 2 teachers assigned => round = 2, status = needs more reading
    - 2 teachers assigned, no 2 scores yet => round = 2, status: waiting for scores
    - 2 scores => round = 2, status: finished, calculate average score





IPO 2016 - Infrastructure

We had a front office with 6 computers (dual monitor setup) to make the evaluation process as quickly as possible.
We had a back office with 2 computers to monitor the evaluation process.
We used 2 copiers to print all papers.

IPO 2016 - Statistics

- 90 student essays
- 25 essays made it to round 2, 65 essays were held back
- 10 essays needed an extra reading in round 1 (score difference >= 3)
- Timing:
  - start reading stage 1: 14.00 h
  - first score round 1: 15.10 h
  - first paper assigned in stage 2: 15.54 h
  - last score: 17.22 h
  - stage 1 including extra reading was completed at 17u02, stage 2 was completed 20 minutes later.